E6.1 <u>Linear congruences</u>

Q7 Find sols to

$$34x \equiv 60 \pmod{98}$$

Sols may or may not exist.
Criterion for existence of solutions is $d \mid 60$ where $d = \gcd(34, 98)$.

$$d = \gcd(34, 98)$$
$$= \gcd(2 \cdot 17, 2 \cdot 7^2)$$
$$= 2$$

and well $2 \mid 60$ ✓. So solutions do exist.
Then by the theorem there are <u>2</u> solution given by

$$x = t + i\frac{98}{2}, \quad i = 0, 1 \pmod{60}$$

$x = t + i49, \; i = 0, 1$
where t is the unique solution to the reduced congruence.

$$17x \equiv 30 \pmod{49}$$

$$\Rightarrow \quad t \equiv 17^{-1} \cdot 30 \pmod{49}.$$

we find $17^{-1} \bmod 49$ by obtaining a Bezout's identity $17n + 49m = 1$, and then $n \equiv 17^{-1} \pmod{49}$, from Euclidean algorithm

$$49 = 2 \cdot 17 + 15$$
$$17 = 1 \cdot 15 + 2$$
$$15 = 7 \cdot 2 + 1$$

then working backwards

$$\begin{aligned} 1 &= 15 - 2 \cdot 7 \\ &= 15 - 7 \cdot (17 - 15) \\ &= 8 \cdot 15 - 7 \cdot 17 \\ &= 8 \cdot (49 - 2 \cdot 17) - 7 \cdot 17 \\ &= \boxed{8 \cdot 49 - 23 \cdot 17} \end{aligned}$$

$$\Rightarrow \quad 17^{-1} \equiv -23 \pmod{49}$$
$$\equiv 26 \pmod{49}$$

$$\Rightarrow \quad t = 17^{-1} \cdot 30 \quad \text{" "}$$
$$= 26 \cdot 30 \quad \text{" "}$$

$$\equiv 780 \pmod{49}$$
$$\equiv 45 \pmod{49}.$$

So the two solutions to $34x \equiv 60 \bmod 98$ are

$$x \equiv 45,\ 45+49 \pmod{98}$$
$$\equiv 45,\ 94. \pmod{98}$$

~~We might have noted~~

$$\cancel{t \equiv -4 \pmod{49}}$$

~~as the generating solutions~~

~~Then the two solutions to the orig. cong would~~

$$x \equiv -4,\ -4+49 \pmod{98}$$
$$\equiv -4,\ 45 \pmod{98}$$

Ex 6.2 Chinese Remainder Theorem.

Q3. Find simultaneous sols to

$$\left\{\begin{array}{l} x \equiv 5 \pmod 6 \\ x \equiv 7 \pmod{11} \\ x \equiv 3 \pmod 5 \end{array}\right\}$$

The moduli 6, 11, 5 are indeed pairwise-coprime so the CRT says there is a unique simultaneous solution

x (modulo $\underbrace{6 \cdot 11 \cdot 5}_{330}$)

Setting up the formulat

$m_1 = 6$, $b_1 = 5$

$m_2 = 11$, $b_2 = 7$

$m_3 = 5$, $b_3 = 3$

$M = 330$, $M_1 = 55$, $M_2 = 30$, $M_3 = 66$

$M_1' \equiv M_1^{-1} \mod m_1$

$\equiv 55^{-1} \mod 6$

$\equiv 1^{-1} \mod 6$

$\equiv 1 \pmod 6$

$M_2' \equiv M_2^{-1} \mod m_2$

$\equiv 30^{-1} \pmod{11}$

$\equiv 8^{-1} \pmod{11}$

$\equiv 7 \pmod{11}$

$M_3' \equiv M_3^{-1} \mod m_3$

$\equiv 66^{-1} \mod 5$

$\equiv 1^{-1} \mod 5 \equiv 1 \pmod 5$

So by the CRT, our solution is

$$x \equiv \sum_{i=1}^{3} b_i M_i M_i' \pmod{M}$$

$$\equiv 5 \cdot 55 \cdot 1 + 7 \cdot 30 \cdot 7 + 3 \cdot 66 \cdot 1 \pmod{330}$$

$$\equiv 293 \pmod{330}$$

Q4. Find smallest number of eggs, x, that might have been in the basket.

$$\cancel{x \equiv 1 \pmod{2}}$$

$$\cancel{x \equiv 2 \pmod{3}}$$

$$x \equiv 3 \pmod{4}$$

$$x \equiv 4 \pmod{5}$$

$$x \equiv 5 \pmod{6}$$

$$x \equiv 0 \pmod{7}.$$

The CRT does not apply straight away.

Note that $x \equiv 5 \mod 6$.

ie $x = 6q + 5$

$$\equiv 3(2q) + 3 + 2$$
$$\equiv 2 \pmod 3$$

So $x \equiv 5 \pmod 6 \Rightarrow x \equiv 2 \pmod 3$

and $x \equiv 5 \pmod 6 \Rightarrow x \equiv 1 \pmod 2$

So consider

$$x \equiv 3 \pmod 4$$

and $x \equiv 5 \pmod 6$

$$\Leftrightarrow \begin{cases} x \equiv -1 \pmod 4 \\ x \equiv -1 \pmod 6 \end{cases}$$

Can we replace this pair with a single congruence).

These will make up a single congruence modulo $\operatorname{lcm}(4,6) = 12$.

It is $x \equiv -1 \pmod{12}$ $\Leftrightarrow$

So our original system is equivalent to (i.e. has the same simultaneous solution as)

$$\left\{\begin{array}{l} x \equiv 11 \pmod{12} \\ x \equiv 4 \pmod{5} \\ x \equiv 0 \pmod{7} \end{array}\right\}$$

The CRT will apply here.